

$$F(x, y, y') = 0$$

y' göre 1. derece bir polinomsa K. derecedendir

$$y' = K \quad (y = Kx)$$

$$F(x, y, p) = 0$$

$$x^2(y')^2 + 2xy' + 5y$$

$$(y')^2 + x^2 = 0$$

$$(y')^2 - x^2 = 0$$

$$F(x, y, p) = 0 \quad [p = y']$$

$$\frac{\partial F}{\partial p} = 0 \quad \text{sistemini çözüyoruz}$$

(x, y) noktalarının kümesinde 1 denklemin
p-diskriminantı çözümleri vardır

$$e^p - p + xy - x - 1 = 0$$

$$e^p - 1 = 0$$

$$e^p = 1$$

$$p = \ln 1$$

$$e^p = \ln 1$$

$$1 + xy - x = 0$$

$$xy = x$$

$$x(y-1) = 0$$

$$y = 1$$

$$p = \ln 1$$

$$x^2 + y^2 + p^2 - 1 = 0$$

$$\begin{cases} 2p = 0 \\ p = 0 \end{cases}$$

$$x^2 + y^2 = 1$$

$$y = \pm \sqrt{1-x^2}$$

$$y \frac{dy}{dx} + x = 0$$

$$\int y dy + \int x dx = 0$$

$$y p^2 + (x-y)p - x = 0 \quad (y \neq 1)$$

$$\frac{y p}{p}$$

$$\frac{x}{-1}$$

$$p = \frac{-x}{y} \quad (p = 1)$$

$$y' = x^3(1-y), y(0) = 3$$

$$dy \cdot \frac{1}{(1-y)} = dx \cdot x^3$$

$$-\ln|1-y| = \frac{x^4}{4} + C$$

$$\ln 2 = C$$

$$\ln|1-y| = -\frac{x^4}{4} + \ln 2$$

$$y = -e^{-\frac{x^4}{4}} + 1$$

$$t^{-1} \frac{dy}{dt} = 2 \cos^2 y, y(0) = \frac{\pi}{4}$$

$$\int \frac{dy}{\cos^2 y} = \int 2t, t^2 = \int \sec^2 y \cdot dy$$

$$C + t^2 = \tan y$$

$$1 + t^2 = \tan y$$

$$\frac{dx}{dt} - x^3 = x$$

$$dx = (x^3 + x) dt$$

$$\int dx \left(\frac{1}{x^3 + x} \right) = \int dt, t = \ln x - \frac{1}{2} \ln|x^2 + 1| + C$$

$$\frac{A}{x} + \frac{Bx+C}{x^2+1} = \frac{1}{x^3+x}$$

$$-\frac{x}{x^2+1}, \frac{x^2+1}{2x} dx = du, -\frac{1}{2} \ln u, -\frac{1}{2} \ln \frac{x^2+1}{2}$$

$$x \left(\frac{dy}{dx} \right) + 2y = x^{-3}$$

$$y' + \frac{2}{x}y = x^{-4}$$

$$e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2$$

$$y \cdot x^2 = x^{-2} \rightarrow -x^{-1} + C$$

$$y = -x^{-3} + Cx^{-2}$$

$$\frac{\cos x}{\cos x} y' + y \frac{\sin x}{\cos x} = 2x \frac{\cos^2 x}{\cos x}$$

$$y' + y \frac{\sin x}{\cos x} = 2x \cos x$$

$$u = e^{\int \frac{\sin x}{\cos x} dx} = e^{-\ln \cos x} = \frac{1}{\cos x}$$

$$- \cos x$$

$$-y \cdot \cos x = 2x \cos x \rightarrow$$

$$-y \cos x = 2x \cos x + C, y = -2x + 2 + C \cos x$$

$$2 \left[x \cdot \cos x - \int \sin x dx \right] = 2x \cos x + 2 \sin x$$

$$\frac{dy}{dx} - \frac{y}{x} = x e^x, y(1) = e - 1$$

$$y' - \frac{y}{x} = x e^x, e^{\int -\frac{1}{x} dx} = e^{-\ln x} = x^{-1}$$

$$\int \left(y \cdot \frac{1}{x} \right)' = \int x e^x, \frac{y}{x} = e^x + C \rightarrow y = x e^x + x C$$

$$x e^x - x$$

$$t^2 \frac{dx}{dt} + 3tx = 1, x(1) = 0$$

$$x' + \frac{3}{t}x = \frac{1}{t^2}$$

$$\int (x + \frac{3}{t})' = \int \frac{1}{t^2}, x + \frac{3}{t} = -\frac{1}{t} + C, x = -\frac{1}{2t} + \frac{1}{2}$$

$$(2x+y)dx + (x-2y)dy = 0$$

$$P(x) = 2x+y, Q(y) = x-2y$$

$$\frac{\partial F}{\partial x} = 2x+y, \frac{\partial F}{\partial y} = x-2y, x^2 + yx - y^2 = C$$

$$0 + x + h'(x) = x - 2y, h(x) = -y^2 + C$$

$$(\cos x \cos y + 2x)dx - (\sin x \sin y + 2y)dy = 0$$

$$P(y) = -\cos x \sin y, Q(x) = -(\cos x \sin y)$$

$$\frac{\partial F}{\partial x} = \cos x \cos y + 2x, \frac{\partial F}{\partial y} = -\sin x \sin y - 2y, \sin x \cos y + x^2 - y^2 = C$$

$$h(x) = -2y, h(x) = -y^2$$

$$\left(\frac{1}{x} + 2y^2x \right) dx + (2yx^2 - \cos y) dy = 0$$

$$P(y) = 4xy, Q(x) = 4yx$$

$$\frac{\partial F}{\partial x} = \frac{1}{x} + 2y^2x, \ln x + y^2x^2 + h(x) = 2y^2x^2 - \cos y, h(x) = -\cos y$$

$$(e^x y + x e^x y) dx + (x e^x + 2) dy = 0, y(0) = -1$$

$$\frac{\partial F}{\partial x} = e^x y + x e^x y, x = u, e^x = \frac{1}{x}, e^x y + x e^x y - e^x y + h'(x) = 0, h(x) = e^x$$

$$\frac{y^2}{2} + \frac{x^2}{2} = C$$

$$\int y^2 dy = y^3 = x$$

$$y - x = C$$

$$y - x - C = 0$$

$$P^3 + yP^2 - x^2y^2P - x^2y^3 = 0$$

$$P^2(P+y) - x^2y^2(P+y) = 0$$

$$(P+y)(P^2 - x^2y^2) = 0$$

$$\frac{dP}{dx} = y \int \frac{dy}{-y} = dx \quad -\ln y = x$$

$$\ln y + x = C$$

$$F(x,y,P) = 0$$

$$P = F(x,y)$$

$$x = g(y,P)$$

$$y = \frac{xP}{2} + \frac{x}{2P} \quad \text{genel u vana ayker}$$

$$f(x,y,P) = 0$$

$$y = xP + f(P) \quad \text{Clairaut denkle}$$

$$P = P + x \frac{dP}{dx} + f'(P) \frac{dP}{dx}$$

$$y = xP - e^P$$

$$P = P + x \frac{dP}{dx} - e^P \frac{dP}{dx}$$

$$(x - e^P) \frac{dP}{dx} = 0$$

$$P = C$$

$$x = e^P$$

$$P = \ln x$$

$$xyP^2 + (x^2 + xy - y^2)P + x^2 + xy = 0$$

$$(xP + xy) \cdot (yP + x) = 0$$

$$\frac{xdy}{dx} = -x - y$$

$$\frac{dy}{dx} = -1 - \frac{y}{x}$$

$$y' + \frac{y}{x} = -1$$

$$e^{\int \frac{1}{x} dx} = x$$

$$y \cdot x = -\frac{x^2}{2} + Cx$$

$$yx + \frac{x^2}{2} = C$$

$$\int y dx = \int x dy$$

$$\frac{y^2}{2} + \frac{x^2}{2} + C = 0$$

$$x^2 + y^2 + C = 0$$

$$y = xP + P^3$$

$$P = P + x \frac{dP}{dx} + 3P^2 \frac{dP}{dx}$$

$$0 = (x + 3P^2) \frac{dP}{dx}$$

$$P = C$$

$$x = -3P^2$$

$$\sqrt{-\frac{x}{3}} = P$$

$$x \leq 0$$

$$P^3 + 3xP - 3y = 0$$

$$y = \frac{1}{3}P^3 + xP \rightarrow \text{clairaut}$$

$$P = P + x \frac{dP}{dx} + P^2 \frac{dP}{dx}$$

$$P = C$$

$$P^2 = x$$

$$(2P^2) (y - xP) = 1$$

$$y - xP = \frac{1}{2P^2}$$

$$y = xP + \frac{1}{2P^2}$$

$$P = C$$

$$x \frac{dP}{dx} = -\frac{1}{P^3} \frac{dP}{dx}$$

$$f(x,y) = 0 \text{ der kleine}$$

$$y = x + \ln x + y \ln x$$

$$y = 2x + \ln x \quad (\text{Differentialrechnung}) \quad (\text{Lagrange})$$

$$p = 2x + \ln x + \frac{1}{p} \frac{dp}{dx}$$

$$-p = 2x \frac{dp}{dx} + \frac{1}{p} \frac{dp}{dx}$$

$$-p = \frac{dp}{dx} (2x + \frac{1}{p})$$

$$\frac{dp}{dx} = 2x + \frac{1}{p}$$

$$x' = -\frac{2}{p}x - \frac{1}{p^2}$$

$$x' + \frac{2}{p}x = -\frac{1}{p^2}$$

$$x \cdot p^2 = -p + C$$

$$y = 2x + \ln p$$

$$x p^2 + p = C$$

$$p(x) =$$

$$y = x p^2 + p^3 \quad \text{perel cirkum}$$

$$y' = x \frac{dp^2}{dx} + p^2 + 3p^2 \frac{dp}{dx}$$

$$(p - p^3) = \frac{dp}{dx} (2x + 3p)$$

$$x' = \frac{2x}{1-p} + \frac{3p}{1-p}$$

$$x' - \frac{2}{1-p}x = \frac{3p}{1-p}$$

$$e = (1-p)^2$$

$$X \cdot (1-p)^2 = \int 3 \cdot (1-p) + 0$$

$$\frac{3p^2 - 6p^3}{3} + 0$$

$$y = x p^2 + p^3$$

$$y = x y' + 2(y')^2 + y$$

$$y' = p$$

$$y = x p + 2p^2 + p$$

$$p y' = p \frac{dx}{dx} + \frac{dp}{dx} (x + 2p + 1) = 0$$

$$0 = \frac{dp}{dx} (x + 2p + 1) = 0$$

$$p = C$$

$$p = -\frac{1}{4}x$$

$$r^2 + 1 = 0$$

$$r^2 = -1$$

$$r = \pm i \quad C_1 \sin x + C_2 \cos x$$

$$x^2 + 2x + 2 = 0$$

$$(x+1)^2 = -1$$

$$r^2 + 5r + 6 = 0$$

$$(r+3)(r+2) = 0$$

$$e^{-3x} + e^{-2x}$$

$$r^2 + 4r + 5 = 0$$

$$(r+2)^2 = -1$$

$$-2 \pm i$$

$$e^{-2x} \sin x + e^{-2x} \cos x$$

$$r^4 - 4r^3 - 14r^2 - 20r + 25 = 0$$

$$\downarrow$$

$$\downarrow$$

$$(r^2 - 2r + 5)^2$$

$$y = x(y')^2 + (y')^2$$

$$y = x p^2 + p^2 \quad \text{Lagrange}$$

$$\frac{dy}{dx} = p^2 + 2p \frac{dp}{dx} x + \frac{dp}{dx}$$

$$(p - p^2) = 2p \frac{dp}{dx} (x + 1)$$

$$p \cdot (1-p) = 2p \frac{dp}{dx} (x+1)$$

$$x' - \frac{2}{1-p}x = \frac{1}{1-p}$$

$$X \cdot (1-p)^2 = \int \frac{1-p}{p^2} + 0$$

$$X = \frac{2p - p^2 + C}{(1-p)^2}$$

$$y = x p^2 + p^2$$

$$r^2 + r' - 1 = 0$$

$$(r+1)(r-1)$$

$$e^{-2x} + e^{+x}$$

$$x e^x \rightarrow (A+B) e^x \cdot x$$

$$(A x^2 + B x) e^x$$

$$\frac{1 + \cos^2 x}{2}$$

$$\frac{1}{2s} + \frac{1}{2} \frac{s}{s^2 + 4}$$

$$\cos x \cdot \frac{\cos^2 x}{1 + \cos^2 x}$$

$$\frac{1}{2} \cos x \cdot \frac{\cos^2 x \cdot \cos x}{2}$$

$$y' - 5y = e^{-5x} \quad y(0) = 0 \quad \text{Laplace}$$

$$sY(s) - \cancel{y(0)} - 5Y(s) = \frac{1}{s-5}$$

$$(s-5)Y(s) = \frac{1}{s-5} \quad \mathcal{L}^{-1}\left(\frac{1}{(s-5)^2}\right)$$

$$Y(s) = \frac{A}{s-5} + \frac{B}{(s-5)^2}$$

$$4s - 5A + B = 1$$

$$A=0 \quad B=1$$

$$\frac{1}{(s-5)^2} = e^{-5x} \cdot x e^{-5x}$$

$$s^2 Y(s) - \cancel{sY(0)} - \cancel{Y'(0)} - \cancel{sY(s)} - \cancel{Y(0)} - 2Y(s) = \frac{4 \cdot 2}{s^3}$$

$$s^2 Y(s) - sY(0) - Y'(0) - sY(s) - Y(0) - 2Y(s)$$

$$s^2 Y(s) - s - 4 - sY(s) - 1 - Y(s) = \frac{8}{s^3}$$

$$(s^2 - s - 2)Y(s) - s - 5 = \frac{8}{s^3}$$

$$\frac{8}{s^3} + s + 5$$

$$\frac{8 + 5s^3 + s^4}{s^3(s^2 - s - 2)}$$

$$s^3(s^2 - s - 2)$$

$$(y'')^2 = 1$$

$$y \sin x dy = \cos x (\sin x - y^2) dx$$

$y^2 = z$

$$y \sin x y' = \cos x - y^2$$

$$y' = \frac{\cos x}{y \sin x} - \frac{y^2 \cos x}{y \sin x} + \frac{y^2 \cos x}{y \sin x} = \frac{\cos x}{y \sin x}$$

$$y y' + y^2 \frac{\cos x}{\sin x} = C$$

$y^2 = z$
 $2y y' = z'$

$$\frac{z'}{2} + z \frac{\cos x}{\sin x} = C$$

$$z' + \frac{2 \cos x}{\sin x} z = C$$

$$4x^2 y'' - 4x y' + 3y = 8x^{4/3} \quad \text{Cauchy Euler}$$

$$4x^2 (D \cdot (D-1)) - 4x (\frac{1}{x} D) + 3y = 8x^{4/3}$$

$$4D - 4D - 4D + 3y = 8x^{4/3}$$

$$4r^2 - 8r + 3 = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{8 \pm \sqrt{64 - 48}}{8} = 2 \parallel$$

$$y'' - 3y' + 2y = 3e^x \cos x \quad y(0) = y'(0) = 0$$

$$s^2 Y(s) - \cancel{y(0)} - \cancel{y'(0)} - 3s Y(s) - 3\cancel{y(0)}$$

$$+ 2 Y(s) = 3 \cdot \frac{(s-1)}{(s-1)^2 + 1}$$

$$(s^2 - 3s + 2) Y(s) = \frac{3s-3}{(s-1)^2 + 1} \rightarrow \frac{A}{(s-2)} + \frac{B}{s-1}$$

$$\frac{3(s-1)}{(s-1)(s-2)(s^2-2s+1)} = \frac{B s + C}{(s-2)(s-1)^2}$$

$$\frac{3(s-1)}{(s-1)(s-2)(s^2-2s+1)}$$

$$y' + z = x$$

$$y(0) = z(0)$$

$$4y + z' = 0$$

$$y' + z = x$$

$$4y + z' = 0$$

$$y(0) = z(0) = 0$$

$$\Delta y + z = x$$

$$4y + z' = 0$$

$$s Y(s) - \cancel{y(0)} + Z(s) = \frac{1}{s}$$

$$4 Y(s) + s Z(s) - \cancel{z(0)} = 0$$

$$\rightarrow s Y(s) + Z(s) = \frac{1}{s}$$

$$2s Y(s) + s Z(s) = 0$$

$$(-s^2 + 4) Y(s) = \frac{1}{s}$$

$$Y(s) = \frac{1}{s(-s^2 + 4)}$$